PrismCheck AI

Round 3 Submission - The Finished product

Team A2Z Creators | Tech Circuit 2026
Live: https://tech-coders.onrender.com/
Source: https://github.com/venkatanarayana15/tech-coders
PDF: https://github.com/venkatanarayana15/tech-coders/blob/main/project_docs/PrismCheck_Round3_Submission.p

A pre-publication brand-governance review workspace. It reviews captions, artwork, PDFs, and DOCX files against a structured brand kit, cites every finding to the exact rule or approved fact, and keeps a human in the approval loop.

1. Executive summary

Product	Deployed web app. Judges only need the live link and the printed demo credentials.
Journey	Sign in -> create a review -> see cited findings -> fix or keep with note -> submit -> review -> approve -> insights.
Three improvements	All three committed improvements are implemented and observable; one is delivered with an honest scope note (see section 5).
Testing	47/47 live end-to-end checks pass on a fresh database; 7 unit tests pass; clean-clone verified.
Verification	Live health check: https://tech-coders.onrender.com/api/health returns 200 with engine status ready.

2. Complete list of features

Access and identity

- Role-based demo sign-in with email and password; all demo credentials are printed on the login page.
- Signed HttpOnly cookie sessions that survive refresh.
- Server-side role enforcement for creator, reviewer, and admin.
- Per-user workspace independence: creators see only their own drafts; reviewers and admins see the shared workspace.

Review workspace

- Creator dashboard with live counts, search, status filters, and priority queues.
- New review from a caption, artwork upload, PDF upload, or DOCX upload.
- Artwork validation for PNG, JPG, and WebP; clear rejection for unreadable files.
- PDF text extraction and review; DOCX text extraction and review.
- PII masking on ingest for emails, phone numbers, and Aadhaar-style IDs.

Review engine checks

- Claims against the approved fact bank (C-01).
- Banned absolute-guarantee review (T-14).
- Urgency and absolute-language review (T-02).
- Preferred-vocabulary review (T-03).
- CIELAB palette-distance review (V-05).
- WCAG contrast and alt-text requirement (A-03).
- Citation Lock: every displayed finding resolves to a real kit item; uncited findings are dropped and counted.

Workflow and decisions

- One-click fixes grounded in approved facts, with capitalization preserved.
- Keep-as-is decision with a required reason.
- Draft lifecycle: draft -> submitted -> approved / changes requested, with backend guards.
- Submit guard blocks drafts with unresolved findings.
- Reviewer approve and request-changes actions with notes.
- Approved drafts are locked against further edits.
- Re-run review against the current kit.
- Revision timeline and audit trail on every draft.

Kit and insights

- Brand Kit reference: 7 rules, 5 approved facts, 5 palette colours, 10 language entries, each addressable by ID.
- Admin Brand Kit editor with automatic version bump and a history trail.
- Insights page: citation coverage, most-flagged rules, most-overridden rules, ambiguous-guideline signal, decision rate, and average review time.
- Clean compliant drafts pass with zero findings.
- Health endpoint and SPA served by the same FastAPI process.
- Seeded demo workspace plus a clean second creator workspace.

3. Technical architecture and implementation

Components

- Frontend: React 18 + Vite 5, handcrafted CSS; production build in frontend/dist is served by FastAPI from the same origin.

- Backend: FastAPI 0.141 on Python 3.14+; SQLModel + SQLite; signed HttpOnly cookie sessions.

- Images: Pillow 12 + numpy 2.5 for CIELAB DeltaE palette checks and WCAG contrast.

- Documents: pypdf 6 for PDFs; zip/XML extraction for DOCX.

- Review engine: deterministic in-process engine by default; optional Gemini 2.5 Flash / Ollama adapters, with Citation Lock enforced on every finding.

- Infrastructure: Render Python web service defined in render.yaml; health check at /api/health.

How components talk

The browser talks only to the FastAPI JSON/multipart API. Sessions are signed HttpOnly cookies. Every route re-checks the user's role on the server, so UI hiding is never the security boundary.

Data flow

A draft is created from caption/alt text/uploads. Text and images are PII-masked, then stored. The deterministic engine runs checks and attaches a source_id to every candidate finding. The Citation Lock resolves each source_id against the kit; findings without a valid source are dropped and counted. Findings and a revision record are persisted, and payloads are returned to the UI.

Models and APIs

Core tables: User, Kit (JSON data), Draft, Finding (JSON fixes and measured values), Revision. Key endpoints: /api/session, /api/users, /api/bootstrap, /api/kit (GET/PATCH), /api/drafts, /api/drafts/{id}/decisions, /submit, /approve, /request-changes, /rerun, /api/insights, /api/files/{name}.

Hardest problem solved

Guaranteeing that every displayed finding is defensible. The Citation Lock is a single server-side choke point: an uncited or unmatched finding can never reach the screen. A close second was keeping fixes correct when edits shift text: decisions now record an explicit note when the exact quote is no longer present, instead of silently doing nothing.

4. Testing details

What we tested and how

- Unit tests (pytest): 7 tests cover overlapping-claim deduplication, approved-fact silence, banned-promise hard stop, urgent-tone soft note, PII masking, and stale-quote replacement behavior.

- Evaluation harness (backend/evaluate.py): 30 synthetic-labelled cases; precision 0.900, recall 0.900, F1 0.900, false-positive rate 0.100.

- Live end-to-end suite (backend/tests/e2e_live.py): 47 checks against a running server on a fresh database - auth, user independence, full caption journey, request-changes loop, image-only and off-brand uploads, real PDF/DOCX extraction, bad inputs, submit/approve guards, role 403s, kit editing, insights, and health/SPA. Result: 47/47 passed.

- Manual browser testing on the deployed link: credentials visible on the login page; admin signs in with email + password; edit-save bumps kit version 4.3 to 4.4; new frontend bundle serves after deploy.

- Clean-clone verification: fresh clone, pip install -r backend/requirements.txt, uvicorn start, / serves the SPA and /api/health returns 200; npm ci and npm run build also succeed.

What we found and fixed

- Image-only uploads were rejected -> fixed in backend and frontend.

- Missing title returned 422 -> now a clean 400.

- Overlapping claim text produced duplicate C-01 findings -> deduplicated.

- A stale quote could silently no-op a fix -> decisions now record a clear note when no text changed.

- Double submit/approve/decide could slip through -> now blocked with 400.

Still known to be imperfect

- Evaluation labels are synthetic, not a real 30-draft dataset.

- No OCR: scanned/image-only PDFs are rejected with a clear message.

- Logo clear-space (L-01) is documented but not pixel-enforced.

- PII masking covers emails, phones, and Aadhaar-style IDs; not a full redaction engine.

- SQLite resets on deploy unless a persistent disk is attached.

- Mobile layout is implemented but not device-tested.

5. The three improvements: committed vs delivered

Improvement 1 - measured evaluation and false-positive reduction

Committed: build a 30-draft labelled set, publish precision/recall, cut false positives toward 10%, and collapse overlapping claim matches into one finding.

Delivered: we delivered the measurement rig and a concrete quality fix. Overlapping claim spans are deduplicated, so '5% cashback for life' now produces one C-01 finding. The harness runs 30 labelled cases and reports precision 0.900, recall 0.900, F1 0.900, and a false-positive rate of 0.100. Honest scope note: the 30 cases are synthetic-labelled, not a collected dataset of 30 real drafts, so these are engineering estimates, not real-world accuracy. The real-world labelled set is our first post-event milestone.

Improvement 2 - editable, versioned Brand Kit and Guideline Health

Committed: turn the read-only kit into an editor for rules, facts, banned terms, preferred vocabulary, and palette; version every change; show which rules are frequently overridden; let an admin promote an override into a kit revision.

Delivered: fully. Admins edit audience, colour tolerance, minimum contrast, rule labels/severity/rule text, approved facts, palette name/hex/role, banned terms, and preferred swaps in-app. Every save bumps the version (verified 4.3 to 4.4) and appends a 10-entry history snapshot. Insights already surfaces most-overridden rules with an ambiguous-guideline signal. Non-admins receive 403 on edit attempts.

Improvement 3 - failure recovery and reliability hardening

Committed: typed, actionable recovery states for empty kits, malformed/oversized uploads, documents with no selectable text, missing citation sources, and model-provider outages; retries/fallbacks; pytest coverage for engine, Citation Lock, role gates, and approval transitions.

Delivered: mostly, with scope honesty. We ship clear 4xx/413 responses for missing title/caption, unreadable or oversized files, plain-text uploads, empty scanned PDFs, duplicate actions, unauthorized access, and approved-draft edits. The model layer falls back to the deterministic local engine when a provider is unavailable. Stale-quote decisions record an explicit note. Double submit/approve/decide are blocked. Engine/decision unit tests plus the 47-check live E2E suite pass. What we did not build: a user-visible retry UI for third-party model outages, because the demo runs the local engine by default and we will not claim a recovery UI we did not ship.

6. Anything else improved after Round 2

- Email + password demo login with credentials printed on the same login page (was one-click-only).
- Added a second demo creator (Rohan) and verified per-user workspace isolation end to end.
- PII masking at ingest for emails, phone numbers, and Aadhaar-style IDs.
- Image-only reviews enabled (previously required a caption).
- Clean 400 for missing title; consistent error messages across upload types.
- render.yaml blueprint and README with clone-and-run instructions.
- Built frontend dist committed so the repo runs without a build step.

7. Product roadmap and future development plan

0-3 months: make the numbers real

Recruit 1-2 mid-size brand teams (fintech or D2C) as pilots. Collect 30 real drafts across captions, banners, and documents; label them with two independent raters; publish precision/recall and inter-rater agreement alongside the synthetic harness results. Success measure: a published, real 30-draft evaluation card.

3-6 months: close the declared gaps

Add OCR for scanned PDFs (Tesseract or a hosted vision model behind the existing Citation Lock). Implement pixel-level logo clear-space enforcement for L-01. Add exportable audit reports (PDF/CSV) so approval records can be handed to compliance.

6-12 months: distribution and integrations

Build Figma and Canva panels so designers are reviewed where they work; add a verified publishing hand-off that carries the approval record to social schedulers/CMS. Add sector compliance packs for regulated industries, starting with financial services.

Who it is for, and what it would take

Brand and content teams at mid-to-large organizations, and the agencies submitting drafts on their behalf. One engineer for OCR and clear-space work, one for integrations, plus pilot customers for labelled data. The roadmap is sequenced so a small team can ship the highest-visibility gaps first, and every item has a measurable exit condition.

8. Honest limitations and disclosures

- The evaluation set is synthetic-labelled, not a real 30-draft dataset.
- No OCR: scanned/image-only PDFs are rejected with a clear message.
- Logo clear-space (L-01) is documented but not pixel-enforced.
- PII masking is not a full redaction engine.
- SQLite is ephemeral on Render unless a persistent disk is attached; the demo workspace reseeds on a fresh disk.
- Mobile layouts are implemented in CSS but not device-tested.

9. Team and declarations

Team	A2Z Creators - Venkata Narayana G V (Team Leader, Developer), Mahesh Babu Ch (Research & Analysis).
Build window	All product code was written during the Tech Circuit 2026 event window.
AI disclosure	This document and the code were prepared with AI assistance. The product uses a deterministic local review engine by default; optional external model adapters are disclosed in the UI when enabled.
Privacy	PII is masked on ingest; demo facts are synthetic; no customer content is used for training.

PrismCheck AI - Round 3 Submission - Team A2Z Creators. Thank you for reviewing.